

SOFTWARE ENGINEERING IN ARTIFICIAL INTELLIGENCE

(21CSE312P)

FITNESS TRACKER USING A.I

Problem Statement:

Regular physical activity is essential for maintaining overall health and fitness. However, individuals often struggle to track their calorie expenditure and optimize workouts based on their body metrics. The **Fitness Tracker** aims to solve this problem by providing a **data-driven solution** that predicts **calories burned** based on user inputs such as **age, gender, BMI, heart rate, and exercise duration**. Additionally, **AI-powered fitness advice** is integrated to provide personalized recommendations, empowering users to make informed decisions about their fitness goals and optimize their exercise routines effectively.

SDG-Aligned Goals:

This project aligns with the **United Nations Sustainable Development Goals (SDGs)**, particularly:

- **SDG 3: Good Health and Well-being** – Encouraging an active lifestyle through personalized fitness tracking and AI-driven recommendations.
- **SDG 9: Industry, Innovation, and Infrastructure** – Utilizing **technology, machine learning, and AI** to enhance fitness tracking solutions.

Dependencies:

The project leverages the following Python libraries:

- **Streamlit** – For building an interactive web-based interface.
- **NumPy** – For numerical computations and data manipulation.
- **Pandas** – For data processing and handling datasets efficiently.
- **Scikit-learn** – For implementing machine learning models to predict calorie expenditure.
- **Requests** – To make API calls for AI-powered fitness advice.

Modules used:

- **User Input Form:** Enter age, BMI, duration, heart rate, body temperature, and gender.
- **Data Preprocessing:** Merges and cleans exercise and calorie datasets.
- **Machine Learning Model:** Predicts calories burned using a trained **Random Forest Regressor**.
- **AI-powered Fitness Advice** – Uses the **Groq API** to provide **personalized AI-generated fitness advice** based on user queries.
- **Insights Section:** Compares user data with historical data and provides fitness insights.

Groq API Integration for Generative AI:

To enhance user experience, the **Groq API** has been integrated to offer **real-time fitness guidance**. Users can input fitness-related queries (e.g., **best exercises, diet tips, recovery methods**) and receive AI-powered recommendations.

Implementation:

1. **API Configuration** – The requests library is used to communicate with the **Groq AI model**.
2. **User Query Handling** – The user submits a fitness-related question, which is sent to the **Groq API endpoint**.
3. **AI Response** – The API returns **personalized fitness advice**, which is displayed on the interface.

GitHub Repository:

- <https://github.com/SANJAYSIVAN/FITNESS-TRACKER>
- <https://github.com/ArpanSurin/Fitness-tracker-app>

Requirements.txt:

streamlit==1.43.0

numpy==1.24.2

pandas==1.5.3

scikit-learn==1.6.1

requests==2.31.0